

Joyce Lai

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SUMMARY

PhD candidate specializing in robust control, optimization, and online learning for high-precision systems. Experienced in designing algorithms with robust stability guarantees under model uncertainty. Demonstrated success translating theory into hardware-in-the-loop performance at Amazon Leo (formerly Project Kuiper).

EDUCATION

University of Michigan, Ann Arbor, MI Aug. 2022 – Present
Doctor of Philosophy, Electrical Engineering

Iowa State University, Ames, IA Aug. 2018 – May 2022
Bachelor of Science (HONS), Electrical Engineering Major, Biomedical Engineering Minor

SKILLS

- **Languages** : MATLAB/Simulink, Python, C, $\text{\LaTeX}$
- **Optimization** : CVX, MPCTools/Casadi
- **Hardware** : Speedgoat Real-Time Target Machine

RESEARCH EXPERIENCE

Peter Seiler Controls Lab Aug. 2022 – Present
Graduate Research Assistant Ann Arbor, MI

- Derived a theoretical robust stability condition for systems with online optimization-based control
- Designed an efficient safety filtering method to enforce robust stability and maintain performance for systems with online optimization-based control
- Demonstrated improved system performance and guaranteed robustness to model uncertainty via online optimization-based control with safety filtering in MATLAB
- Developed a performance analysis framework for systems with bandwidth and robustness constraints to understand when advanced control offers significant improvements given limited sensor data

Amazon Leo (formerly Project Kuiper) May 2025 – Aug. 2025
Applied Scientist Intern II (L5) Northridge, CA

- Conducted a performance-robustness tradeoff study between online gradient descent and recursive least squares algorithms for fast steering mirror jitter suppression in MATLAB
- Conducted real-time, hardware-in-the-loop (HIL) testing of both online learning-based control algorithms on a Speedgoat performance real-time target machine
- Demonstrated 75% jitter reduction during real-time, HIL testing via online learning-based control with exponential computational cost reduction

Laboratory of Integrated Optical Sensors (LIOS) Jan. 2021 – May 2022
Undergraduate Research Assistant Ames, IA

- Developed a paper-based miRNA biosensor with rapid detection and high sensitivity for point-of-care applications

Robotic Swarm Control Lab May 2020 – May 2021
Undergraduate Research Assistant Houston, TX

- Developed a computational fluid dynamics (CFD)-based propulsive force model for milliscale medical robots via the penalty method of solving the Navier-Stokes equations for incompressible fluids
- Developed an automated CFD-based design optimization pipeline for milliscale medical robots

Nilakanta Lab

Undergraduate Research Assistant

Jan. 2019 – May 2020

Ames, IA

- Performed data pre-processing for ML-based biomarker detection of Parkinson's disease
- Led the undergraduate student research team and delegated project tasks

TECHNICAL PUBLICATIONS & PRESENTATIONS

1. *(In Review)* **J. Lai** & P. Seiler, "Fundamental Disturbance Rejection Limits for LTI Systems," *IEEE Control Systems Letters*, 2026.
2. **J. Lai** & P. Seiler, "Safety Filter for Robust Disturbance Rejection via Online Optimization," *European Journal of Control*, 2025, DOI: 10.1016/j.ejcon.2025.101337.
3. **J. Lai** & P. Seiler, "Robust Online Convex Optimization for Disturbance Rejection," *2024 IEEE 63rd Conference on Decision and Control (CDC)*, Milan, Italy, 2024, pp.8876-8883, DOI: 10.1109/CDC56724.2024.10886354.
4. J. Qian, Q. Zhang, **J. C. Lai**, Y. Wang and M. Lu, "Rapid MicroRNA Detection Using Paper-Based Isothermal Amplification," *2021 IEEE Sensors*, Virtual, 2021, pp. 1-4, DOI: 10.1109/SENSORS47087.2021.9639797.
5. **J. Lai**, "Portable Biosensors: Fluorescence-based MicroRNA Quantification for an Affordable, Compact, Disease Detection Device," ISU 15th Annual Symposium on Undergraduate Research and Creative Expression, United States, Iowa, Ames, April, 2021.
6. **J. Lai**, "Automated Design Optimization of Rotating Magnetic Swimmers for Minimally Invasive Surgeries," ISU 15th Annual Symposium on Undergraduate Research and Creative Expression, United States, Iowa, Ames, April, 2021.
7. **J. Lai**, "Development of a Computational Fluid Dynamics Force Model for a Rotating Magnetic Swimmer Able to Remove Blood Clots," University of Houston Neurotechnologies to Help the Body Move, Heal and Feel Again Research Experience for Undergraduates Poster Presentation, United States, Texas, Houston, July, 2020.
8. **J. Lai**, "Identification of Novel Biomarkers using a Big Data approach: Integrating Brain Image Data with AD/PD Databases," Iowa State University Honors Poster Presentation, United States, Iowa, Ames, May, 2019.

PROJECTS

Automatic Control Systems

Fall 2021

Proportional, Integral, and Derivative (PID) Controller for Water Reservoir System

Ames, IA

- Designed a PID controller for a water reservoir system
- Demonstrated improved system stability and performance via PID control in MATLAB and analytically

Communication Systems I

Fall 2020

Code Division Multiple Access (CDMA) Effect on Jamming

Ames, IA

- Designed a CDMA modulation scheme for a wireless communication system
- Demonstrated improved system robustness to jamming via CDMA modulation scheme in MATLAB and Simulink

TEACHING EXPERIENCE

Department of Electrical and Computer Engineering (ECE)

Aug. 2021 – May 2022

Signals & Systems II (EE 324) Teaching Assistant

Ames, IA

- Re-designed existing course laboratory curriculum and graded laboratory assignments/reports
- Led laboratory sessions involving signal processing and system modeling and analysis in MATLAB and Simulink

Department of Electrical and Computer Engineering (ECE)

Aug. 2019 – Jul. 2020

Introductory C-Programming (CprE/SE 185) Teaching Assistant

Ames, IA

- Designed new team-based Learning (TBL) course curriculum and graded exams
- Led supplementary recitations and exam review sessions

Academic Success Center (ASC)

Feb. 2019 – May 2019

Calculus II (MATH 166) Tutor

Ames, IA

- Designed tutoring curriculum and led small group tutoring sessions

HONORS & AWARDS

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| • Rackham Student Government (RSG) Empowerment Award | Spring 2025 |
| • Spring 2022 Student Marshal for the College of Engineering | Spring 2022 |
| • Iowa State University Professional and Scientific Team Cytation Award | Spring 2022 |
| • Highest 2% Engineering Sophomore | Spring 2019 |
| • IEEE Eta Kappa Nu (IEEE-HKN) Honor Society | Spring 2019 |
| • Cima Scholarship in Electrical and Computer Engineering | Spring 2019 |
| • Iowa State University Honors Research Grant | Spring 2019 |

VOLUNTEER, LEADERSHIP & ACTIVITIES

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| • Senate Advisory Committee on University Affairs, Student Representative | Sep. 2025 – Present |
| • Honorary Degree Committee, Student Representative | July 2025 – Present |
| • Rackham Student Government (RSG), Chief of Staff | May 2025 – Present |
| • Rackham Student Government (RSG), Executive Board | May 2024 – May 2025 |
| • Graduate Student Advisory Council (GSAC), Board | Mar. 2024 – May 2025 |
| • Rackham Student Government (RSG), Board | Jan. 2024 – May 2024 |
| • Movement for Underrepresented Sisters in Engineering & Science, Executive Board | Jan. 2024 – May 2024 |
| • ECE Graduate Student Council (GSC), Board | Jan. 2023 – Present |
| • International Student Council (ISC), Student Advocacy Committee Director | Apr. 2019 – May 2022 |
| • IEEE Eta Kappa Nu (IEEE-HKN), Service Chair & Help-Room | Apr. 2019 – Dec. 2019 |
| • Iowa State Leadership Experience Conference Attendee | Spring 2019, Spring 2020 |
| • Science Bowl Volunteer | Feb. 2019 |
| • Engineering Career Fair Ambassador | Feb. 2019 |
| • ISU Symphony & Chamber Orchestra | Aug. 2018 – May 2019 |